

8. Interdisciplinary Modeling in Quantitative Trading (Part 3/3)

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C O N T E N T S

1. Predictability in Finance

Foundations of Finance Theory

Theoretical Frameworks in Finance and their Applications

A Conjectural Mapping among Theories and Strategies

2. Chaos Theory

Chaos, Randomness, Stochastic

Logistic Map

3. Predictability in Physics

Mean-squared Displacement

0-1 Chaotic Test

Entropy

1. Predictability in Finance: Foundations of Finance

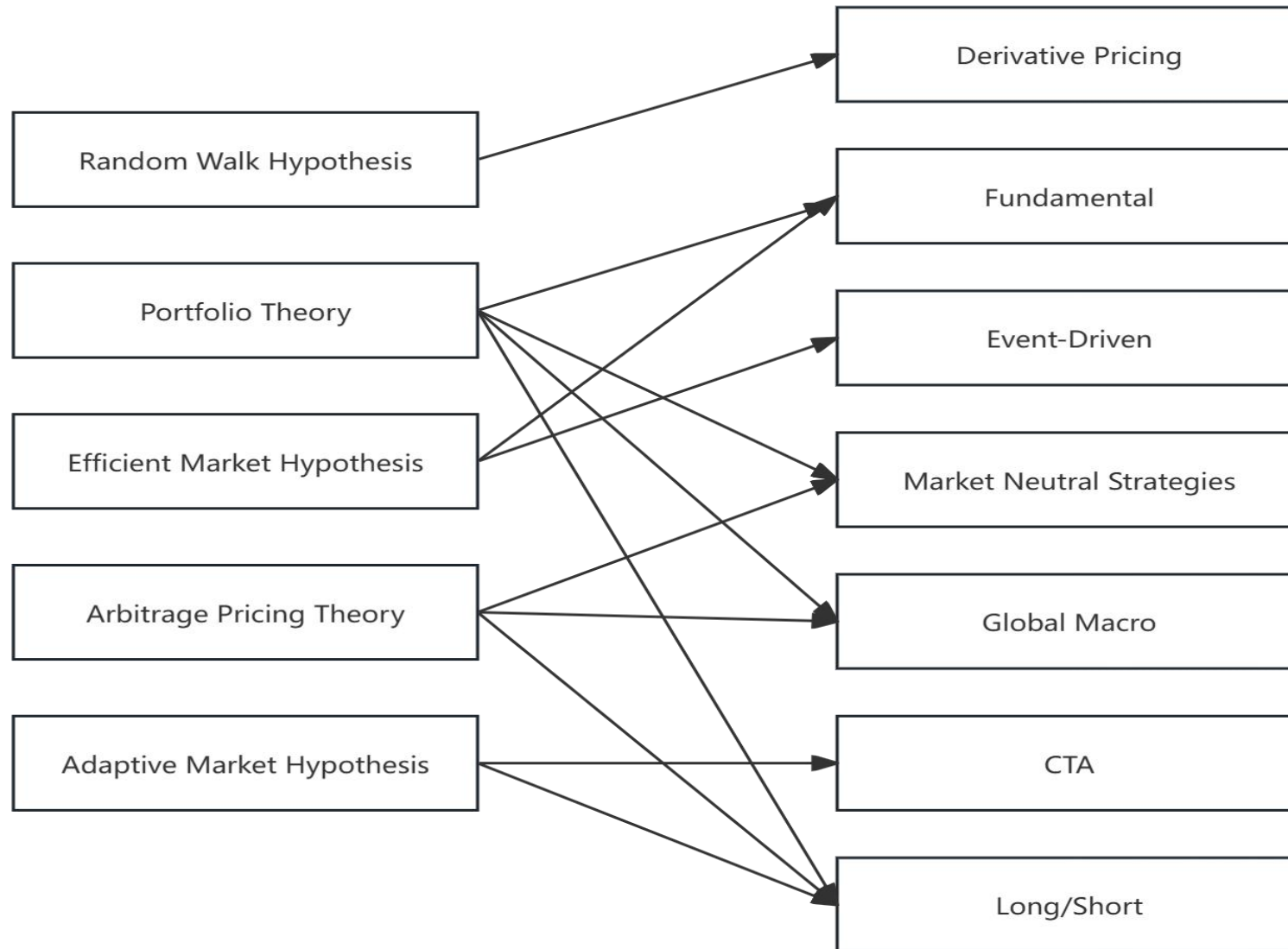
Theory Name	Main Contributor	Time	Representative Paper
Random Walk Hypothesis (RWH)	Louis Bachelier (Early Proposer)	1900	"The Theory of Speculation"
	Paul Cootner (Systematic Exposition)	1964	"The Random Character of Stock Market Prices"
Modern Portfolio Theory (MPT)	Harry Markowitz	1952	"Portfolio Selection"
Efficient Market Hypothesis (EMH)	Eugene Fama	1970	"Efficient Capital Markets: A Review of Theory and Empirical Work"
Arbitrage Pricing Theory (APT)	Stephen Ross	1976	"The Arbitrage Theory of Capital Asset Pricing"
Adaptive Market Hypothesis (AMH)	Andrew Lo	2004	"The Adaptive Markets Hypothesis: Market Efficiency from an Evolutionary Perspective"

1. Predictability in Finance: Theories in Finance and their Applications

Major Hypothesis and Theories in Finance	Pricing	Optimization	Attribution	Prediction
Random Walk Hypothesis (RWH)	Black-Scholes (Option Price) Vasicek model (Interest) HJM model (Interest)	—	Underlying asset's price (Delta and Gamma), time (Theta), volatility (Vega), and interest rates (Rho)	—
Modern Portfolio Theory (MPT)	—	Mean-Variance Analysis	μ, σ	—
Efficient Market Hypothesis (EMH)	CAPM	—	α, β	—
Arbitrage Pricing Theory (APT)	Multi-Factor Pricing Models	Factor Exposure Optimization	Multi-Factor Attribution	Factor Return Forecasting
Adaptive Market Hypothesis (AMH)	Nonlinear Pricing Models (Not Yet)	State-Dependent Portfolio Allocation (Not Yet)	Regime-based Attribution (Not Yet)	Machine Learning (Not Yet)

Current research on the Adaptive Markets Hypothesis (AMH) is largely driven by empirical tests of its validity, rather than by the construction of new models from its foundations.

1. Predictability in Finance: A Conjectural Mapping among Theories and Strategies



2. Chaos Theory: What is Chaos?

Dimension	Randomness	Stochasticity	Chaos
Definition	The outcome cannot be predicted given all known information and laws.	Future states are not determined solely by its past but are also governed by probabilistic laws.	Chaos is apparently random, unpredictable behavior arising from a deterministic, non-linear system.
Determinism vs. Randomness	Purely non-deterministic.	The model is deterministic.	The model is deterministic, with no random input, providing random like output.
Predictability	Statistically predictable (Probability distributions, expectations, etc.)	Statistically predictable	Unpredictable in the long run/ Periodically predictable
Sensitivity to Initial Conditions	Insensitive (events are independent).	Model-dependent (e.g., a random walk is sensitive).	Extremely sensitive (Butterfly Effect).
Core Models	Monte Carlo Simulation	ARMA model	Lorenz System

2. Chaos Theory: Logistic Map

$$x_{n+1} = \lambda x_n(1 - x_n), \forall x \in [0,1]$$

In Biology,

x_n is the proportion of the population that lives at time n ;

$1 - x_n$ is the proportion of death;

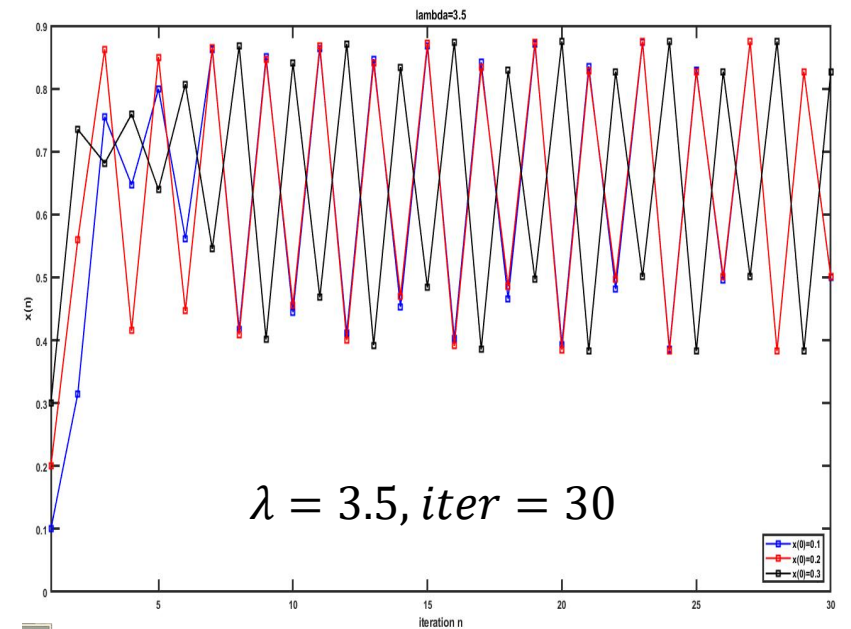
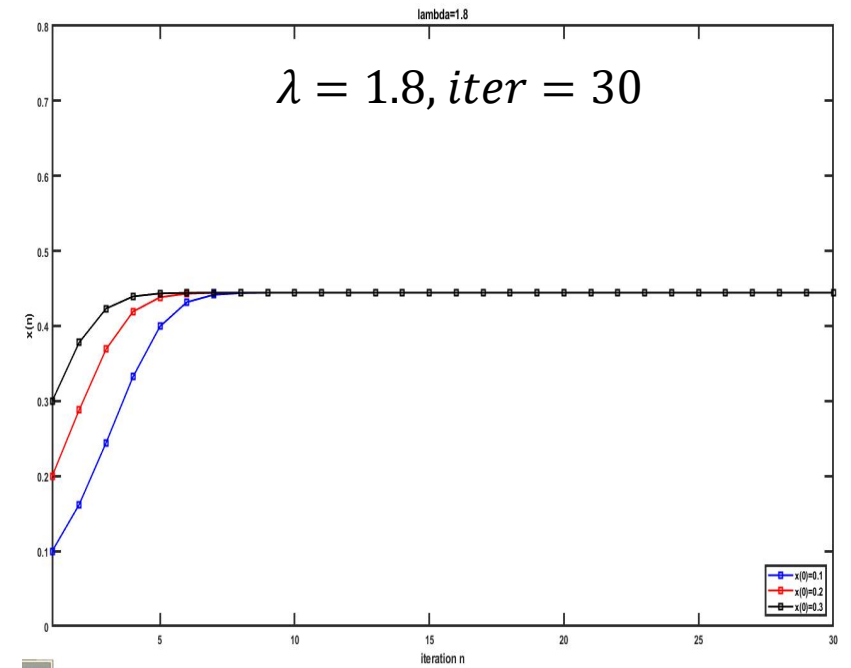
λ is the reproduction/growth rate of the living population.

In Finance,

x_n is the proportion of shares holding by investors at n

$1 - x_n$ is the proportion of shares sold by holders.

λ is the growth rate of the current stock holders buying more shares.

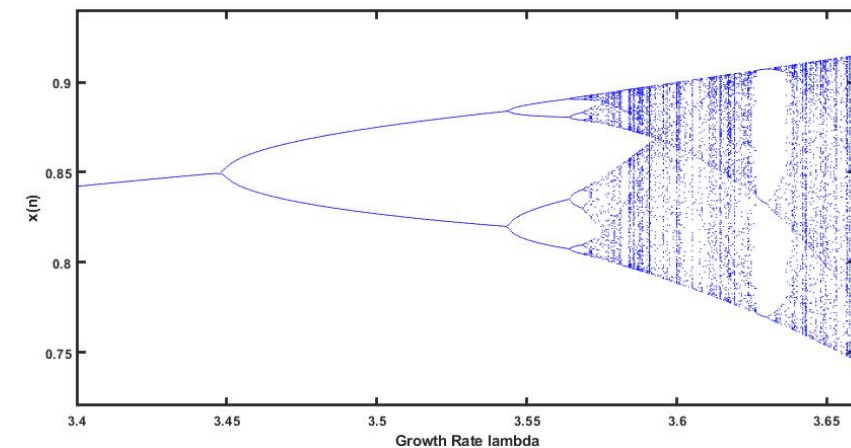
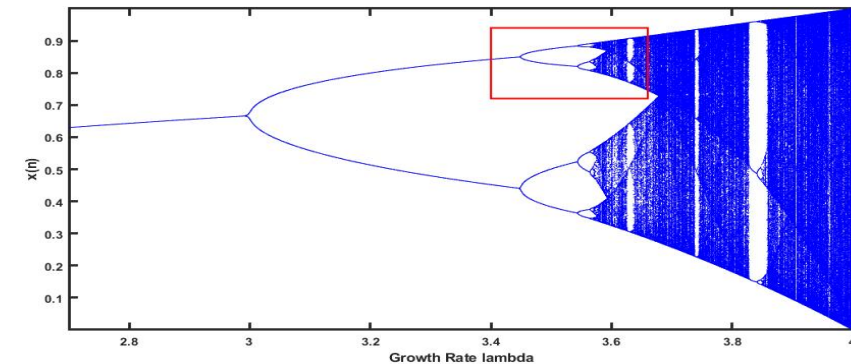
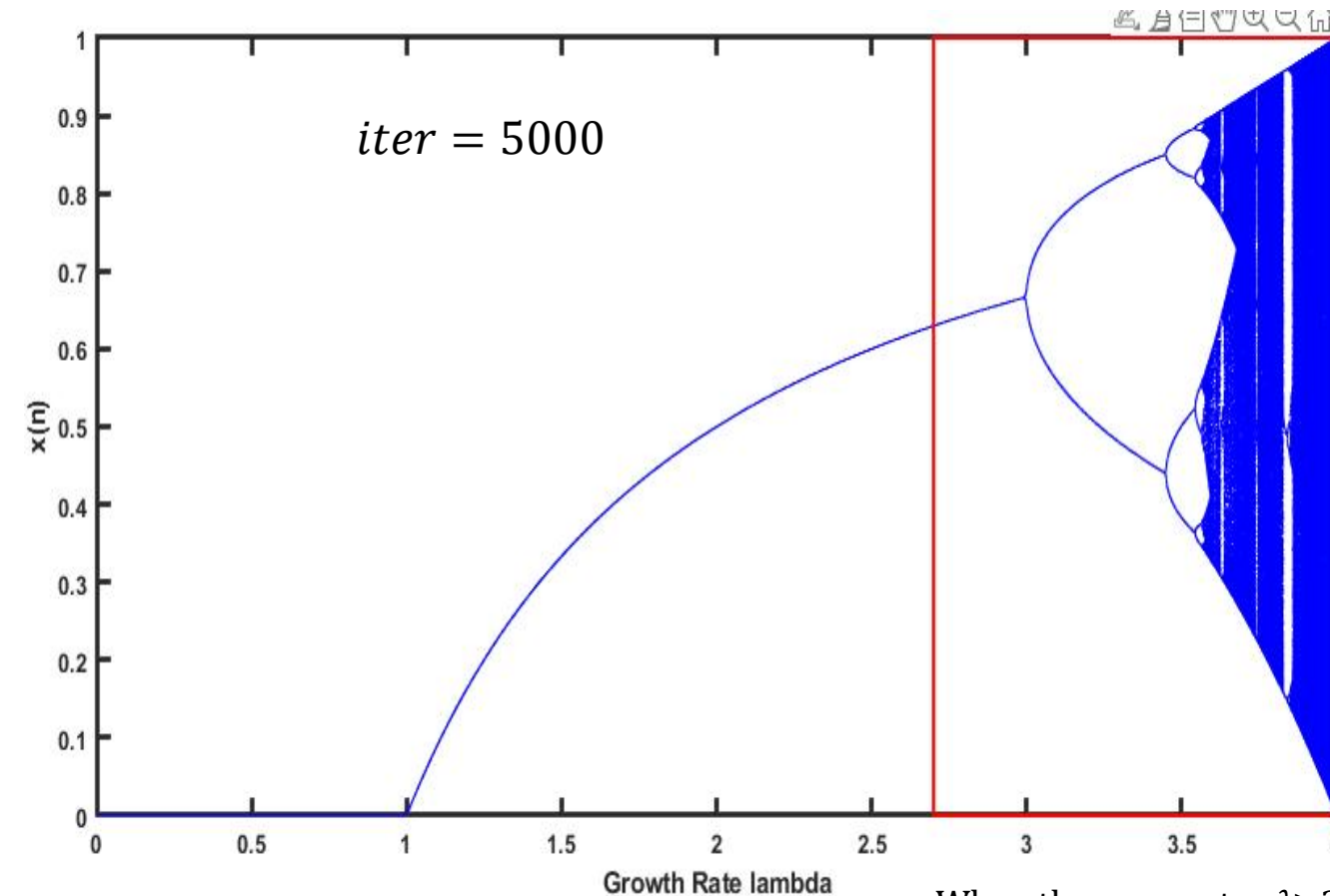


2. Chaos Theory: Logistic Map

Chaos is a complex and irregular behavior that occurs despite the fact that the ODE describing the dynamic system has no probabilistic ambiguity and the next state is completely and uniquely determined.

Self-similarity is a property where a part of an object or process is similar to the whole, appearing the same at different scales.

Butterfly Effect: The flap of a butterfly's wings in Brazil sets off a tornado in Texas. (Sensitivity to initial conditions)

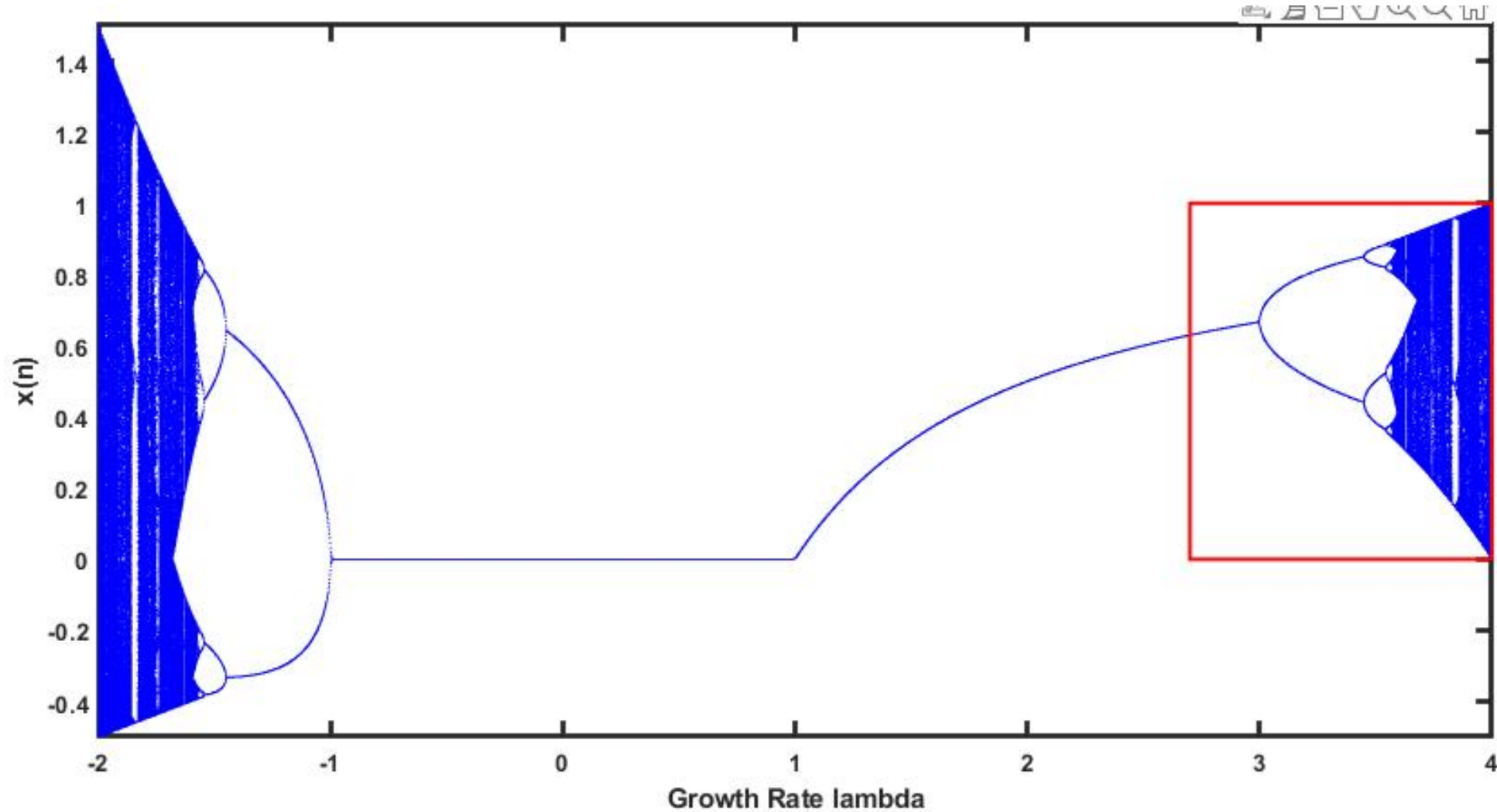


When the parameter $\lambda > 3.56994\dots$, the logistic map exhibits chaotic behavior.

2. Chaos Theory: Logistic Map

Conjecture:

A negative growth rate of Logistic map will create fat-tails?



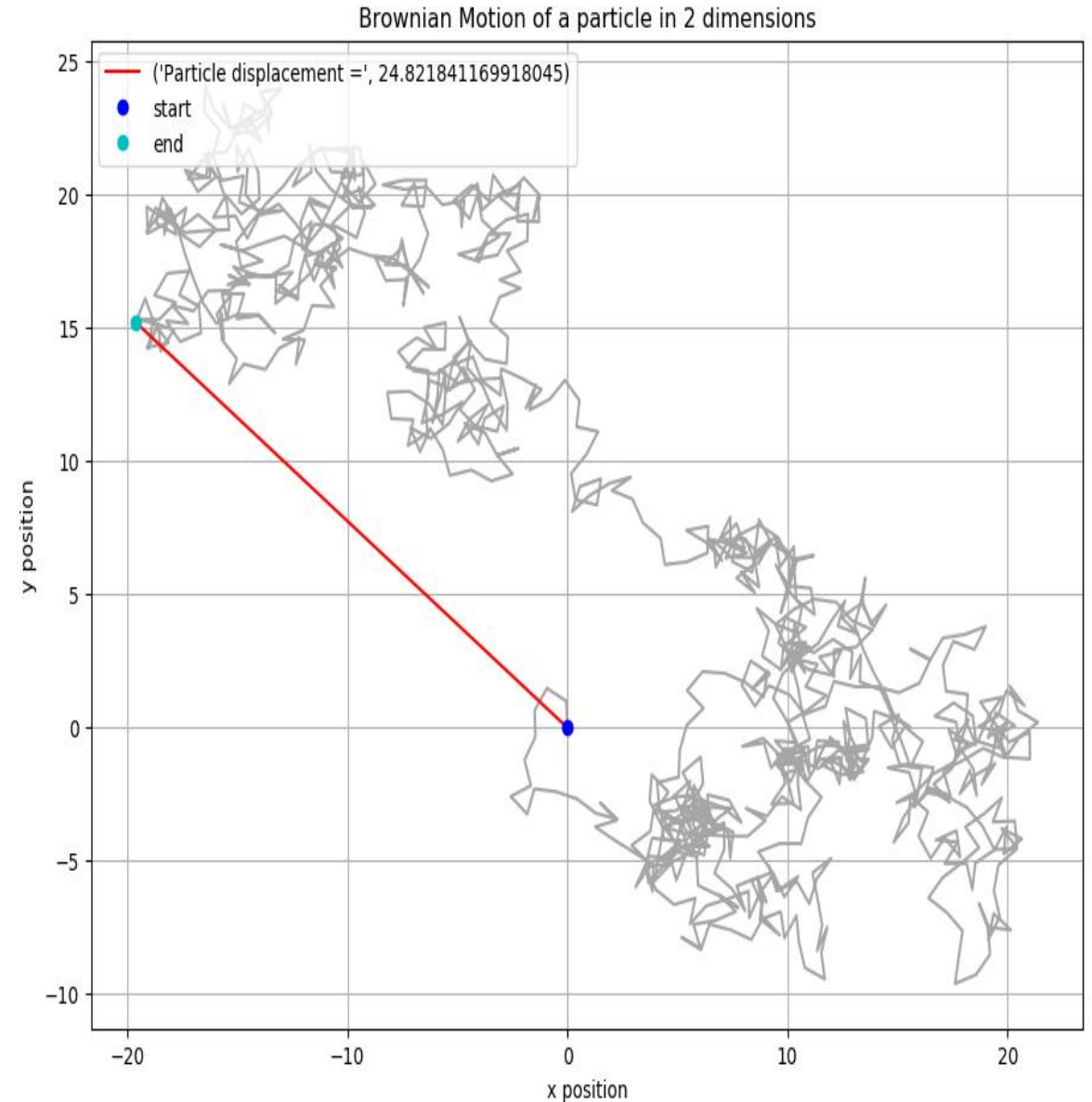
3. Predictability in Physics: 0-1 Chaotic Test (Mean-squared Displacement)

In statistical mechanics, the mean squared displacement (MSD), also called mean square displacement, average squared displacement, or mean square fluctuation, is a measure of the deviation of the position of a particle with respect to a reference position over time.

The MSD at time t is defined as an ensemble average:

$$\text{MSD} \equiv \langle |\mathbf{x}(t) - \mathbf{x}_0|^2 \rangle = \frac{1}{N} \sum_{i=1}^N |\mathbf{x}^{(i)}(t) - \mathbf{x}^{(i)}(0)|^2$$

where N is the number of particles to be averaged, $x^{(i)}(t)$ is the position of the i-th particle at time t, $x^{(i)}(0)$ is the reference position of the i-th particle.



source: <https://stackoverflow.com/questions/59020274/mean-square-displacement-as-a-function-of-time-in-python>

3. Predictability in Physics: 0-1 Chaotic Test

1. Given an observation $\varphi(j)$ for $j = 1, \dots, N$:

For $c \in (0, \pi)$, we compute the translation variables

$$p_c(n) = \sum_{j=1}^n \varphi(j) \cos(jc)$$

$$q_c(n) = \sum_{j=1}^n \varphi(j) \sin(jc)$$

for $n = 1, 2, \dots, N$.

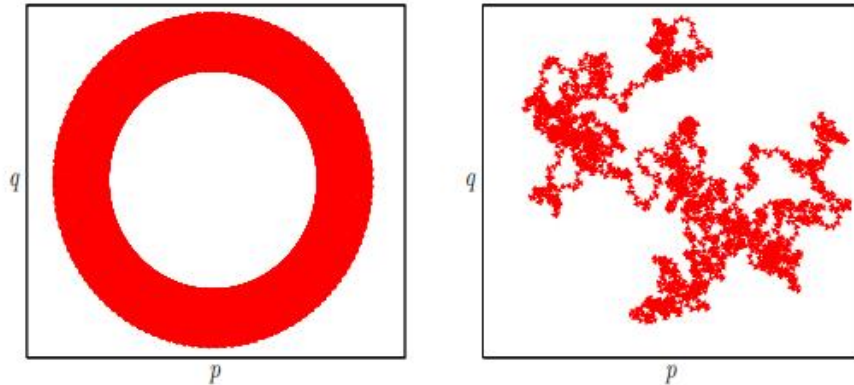


Figure 1: Plot of p versus q for the logistic map $x_{n+1} = \mu x_n(1 - x_n)$. Left: Regular dynamics at $\mu = 3.55$; Right: Chaotic dynamics at $\mu = 3.9$. We used 5000 data points.

2. The diffusive (or non-diffusive) behaviour of p_c and q_c can be investigated by analyzing the mean square displacement $M_c(n)$.

$$M_c(n) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N ((p_c(j+n) - p_c(j))^2 + (q_c(j+n) - q_c(j))^2)$$

3. We then compute the asymptotic growth rate K_c of the mean square displacement.

$$K_c = \text{corr}(\xi, \Delta)$$

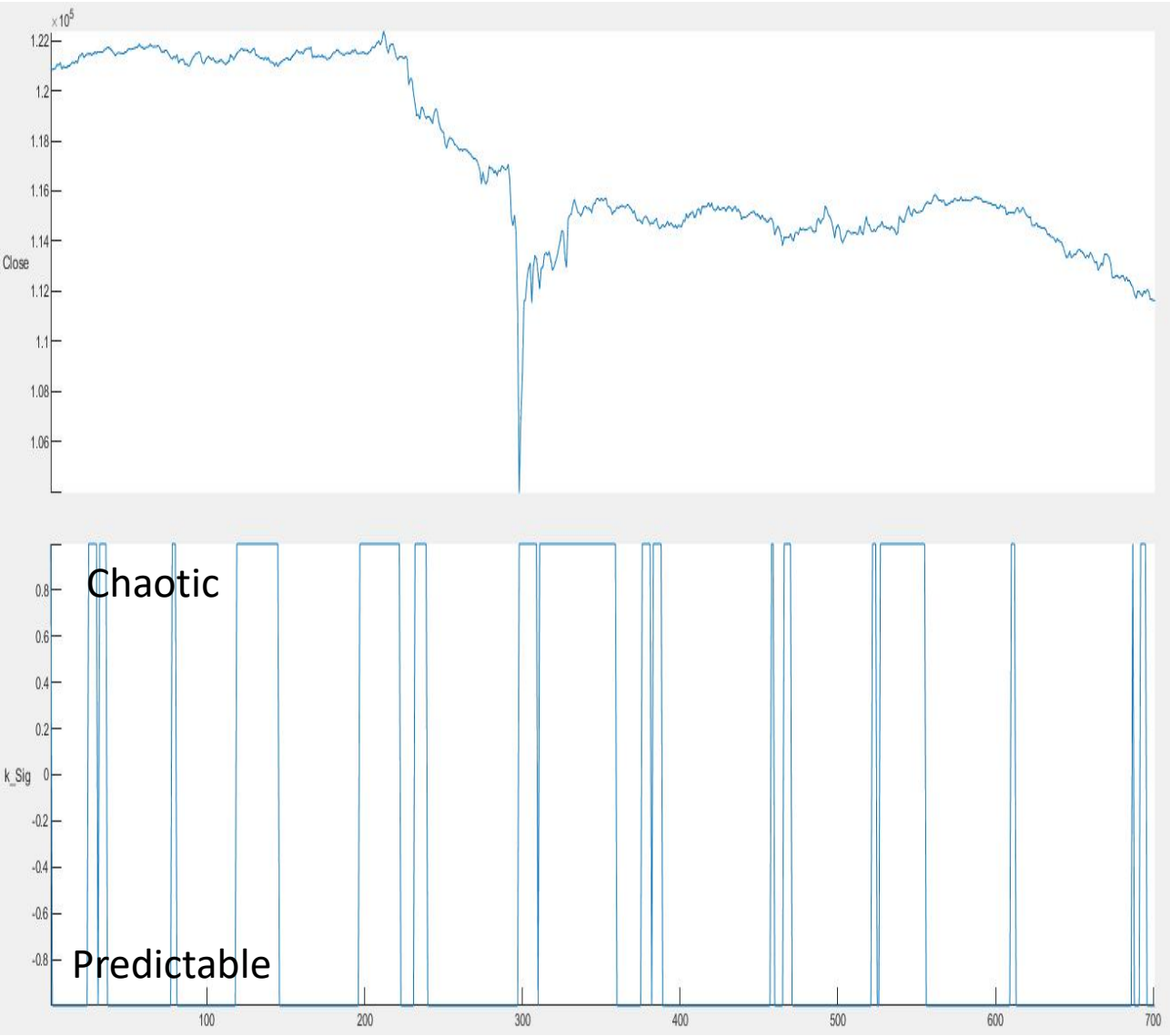
where $\xi = \{1, 2, 3, \dots, n_{\text{cut}}\}$, $\Delta = \{M_c(1), M_c(2), \dots, M_c(n_{\text{cut}})\}$.

4. Steps 1–3 are performed for N_c values of c chosen randomly in the interval $(0, \pi)$. In practice, $N_c = 100$ is sufficient. We then compute the median of these N_c values of K_c to compute the final result $K = \text{median}(K_c)$. Our test states that a value of $K \approx 0$ indicates regular dynamics, and $K \approx 1$ indicates chaotic dynamics.

In practice, c is uniformly random in $(\pi/5, 4/5\pi)$

- Gottwald, G. A., & Melbourne, I. (2009). On the implementation of the 0–1 test for chaos. *SIAM Journal on Applied Dynamical Systems*, 8(1), 129-145.
- Gottwald, G. A., & Melbourne, I. (2004). A new test for chaos in deterministic systems. *Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences*, 460(2042), 603-611.

3. Predictability in Physics: Empirical Results of 0-1 Chaotic Test



Descriptive Statistics	Mean	Standard Deviation	Skewness	Kurtosis	Cumulative Return
Chaotic	-0.0006%	0.0023	-5.8577	260.2455	-1.99%
Non-chaotic	-0.0043%	0.0016	-0.4747	24.5646	-29.22%

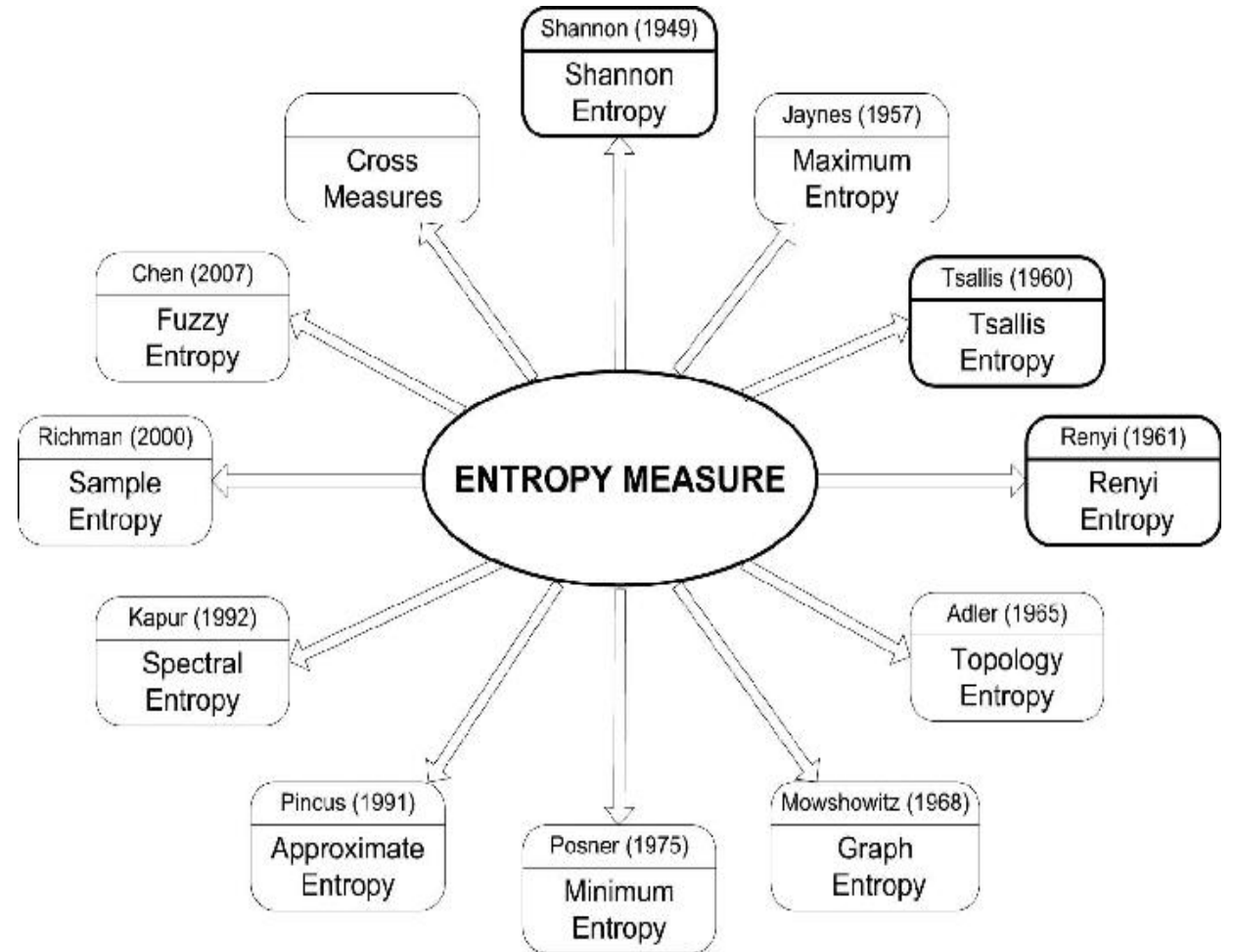
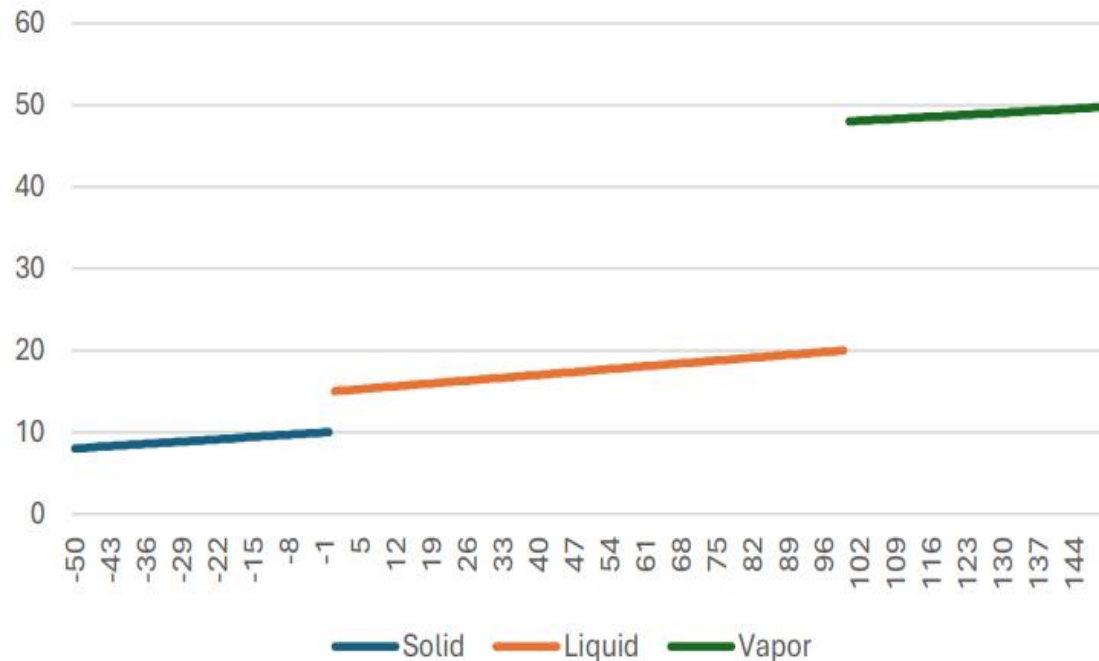


3. Predictability in Physics: Entropy, the Concept

The (Shannon) Entropy of $\mathbf{p}$ is

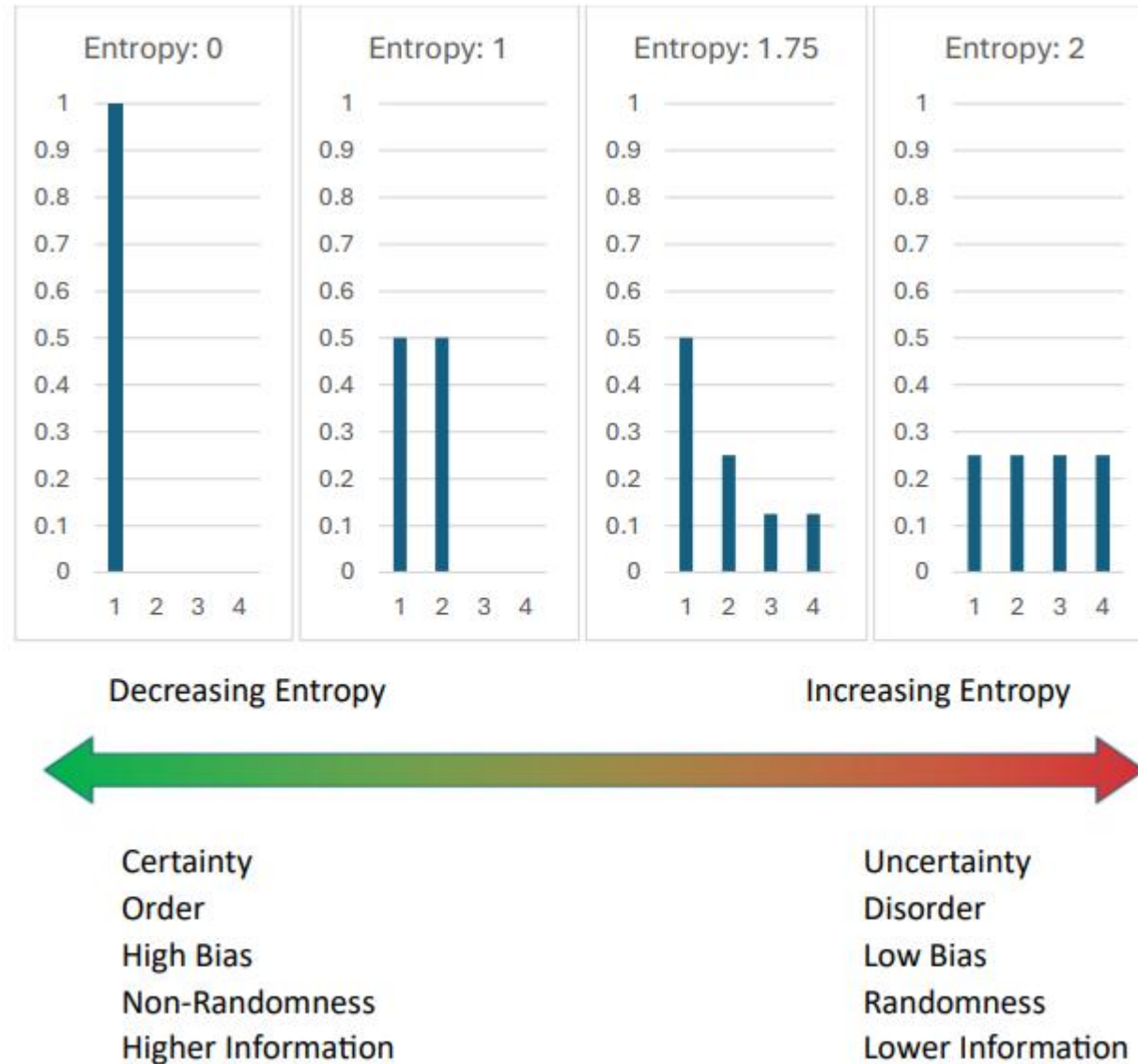
$$H(\mathbf{p}) = \sum_{i=1}^n p_i \log\left(\frac{1}{p_i}\right)$$

Entropy of the states of water

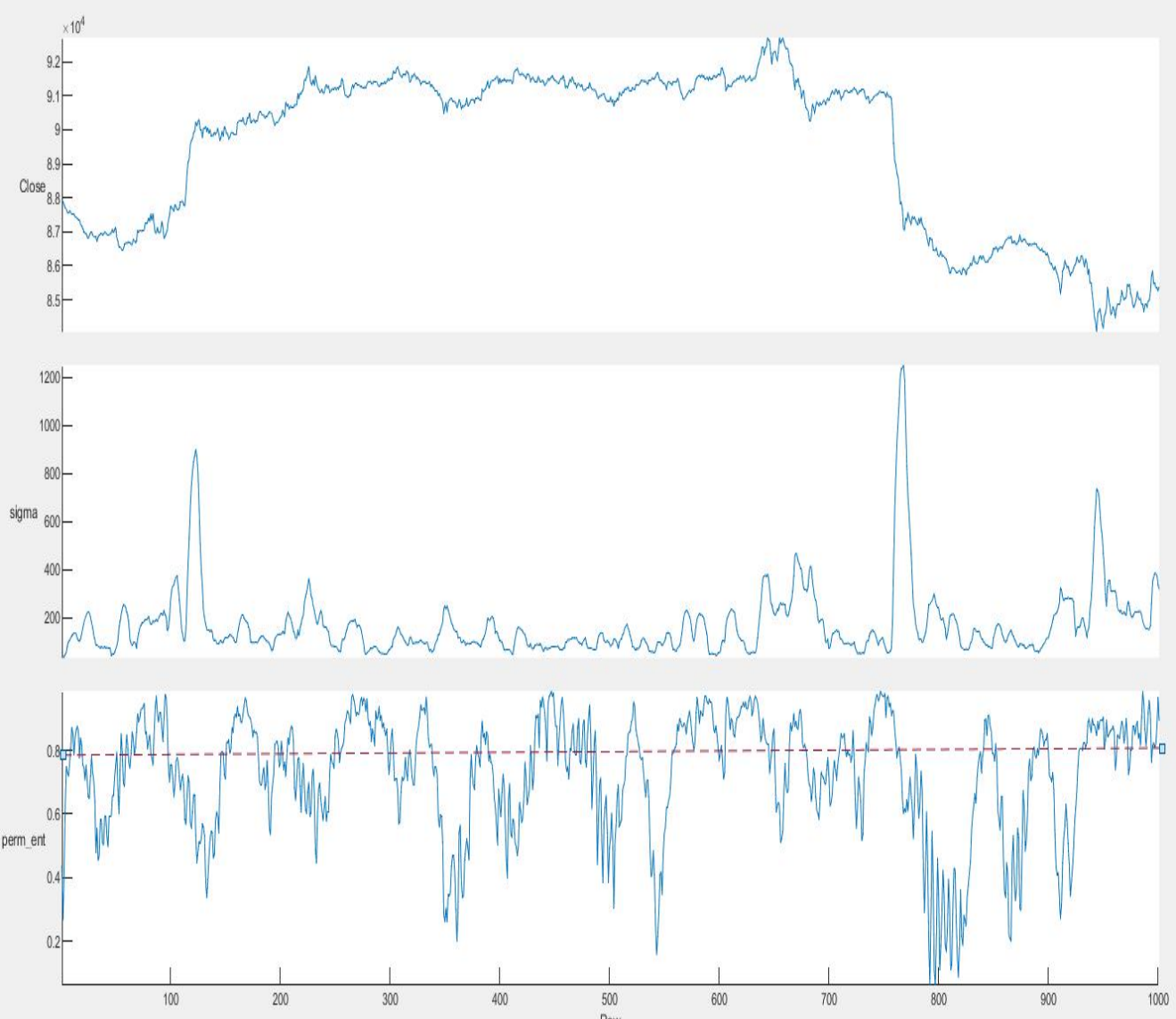


- Borowska, M. (2016). Entropy-based algorithms in the analysis of biomedical signals. *Studies in Logic, Grammar and Rhetoric*, 43(1), 21-32.

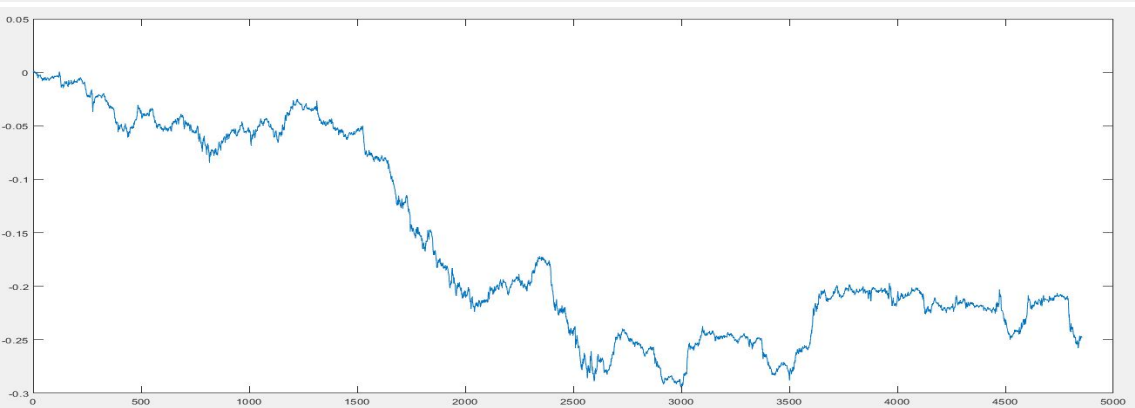
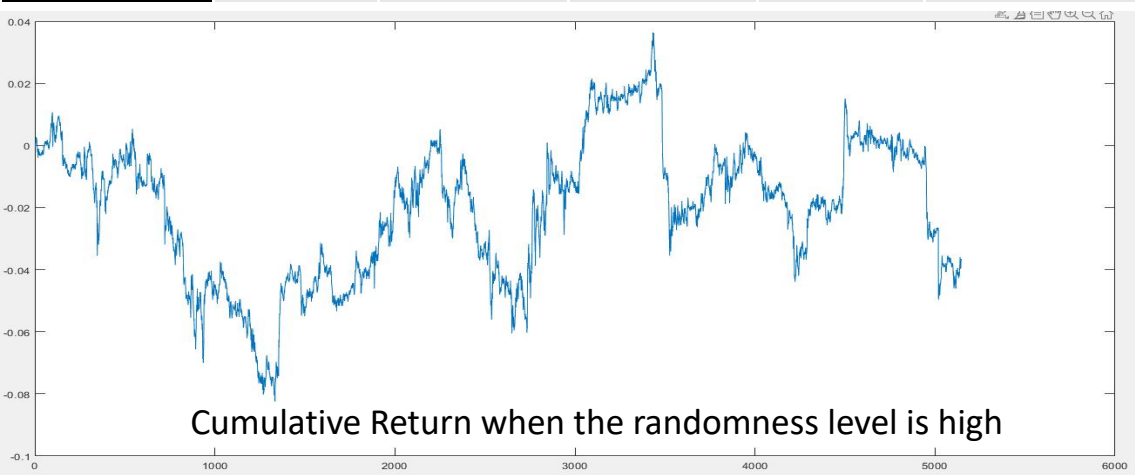
3. Predictability in Physics: Entropy, Interpretation



3. Predictability in Physics: Entropy, Empirical Application



Descriptive Statistics	Mean	Standard Deviation	Skewness	Kurtosis	Cumulative Return
Chaotic	-0.0007%	0.0017	-0.2350	17.7241	-3.64%
Non-chaotic	-0.0051%	0.0018	-0.2123	9.2760	-24.78%



Assignment 4 & 5: Backtesting and Improvement

Assignment 4 (Backtesting)

1(100). Backtest the the strategy you've designed. And fill Table 3 (Empirical Test).

Assignment 5 (Improvement)

1(80). Fill Table 4 (Evaluation of Empirical Results).

2(10). Describe its performance in scenarios (bull, bear market conditions). Illustrate the pros and cons of your strategy.

3(10). Figure out some way to improve your strategy.

- If your strategy doesn't perform well in the backtest, that's completely fine. Please submit the assignment on time, and we will work on understanding and improving it together. It's very common for a strategy to turn out not to work. Your grade is based on the effort you put into each step, not on the strategy's performance.
- If your strategy turns out to perform well, you'll receive bonus points in your class assessment.

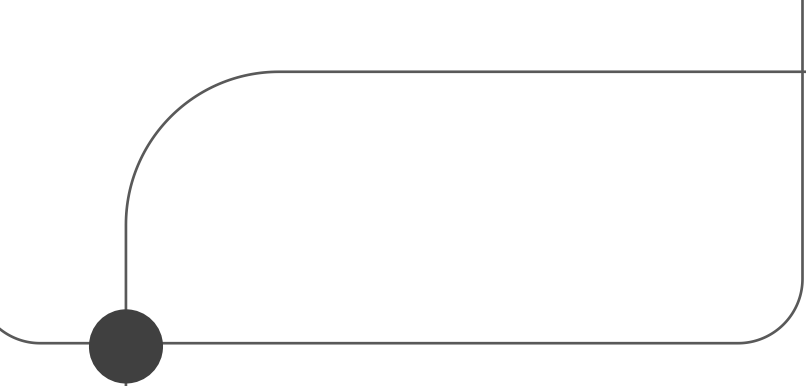
Appendix

Empirical Test (Back Test, Forward Test, Paper Trading, Live Trading)	
Benchmark	
Performance Assessment	Sharpe Ratio, Calmar Ratio, Maximum Drawdown, Expected Return (Annualized, Compounding/Non-compounding), Trading Frequency, Average Holding Period, Winning Probability, etc.
Back Test 1	The back-testing period 1 should contain both bull and bear market conditions.
Back Test 2	The back-testing period 2 should be the nearest 3-6 months to now.
Back Test Scenario 1	Test the strategy performance during typical risk events such as the financial crisis in 2008, trade war, etc.
Back Test Scenario 2	
Paper Trading (Forward Test)	Paper trading is a critical step in strategy development. However, it excludes real-world frictions and operational risks inherent to live markets, such as slippage, market impact, latency, partial order fills, and technical disruptions (e.g., internet disconnections or data feed failures).
Live Trading	
Capacity Test	

Table 3. Empirical Test

Evaluation of Empirical Results	
Refine the Strategy	● Does the strategy’s return, risk, capacity, and trading frequency align with expectations? ● What caused the drawdowns, and are there methods to mitigate them? ● Would earlier or later entry/exit timing improve performance? ● Can risk be reduced or trading frequency increased while maintaining target returns? ● Can the strategy be extended to other instruments or markets to expand opportunities?
Applicable Market Conditions	Observe the best performing market conditions of the strategy. (e.g., high volatility, uptrend with low volume, etc.)
Risk/Position Management	Position control, stoploss/stopgain, hedging, risk redflags (liquidity risk, systemic risk, etc.)
Execution Risk Assessment	Assess the impact of order fill rate, execution latency, slippage, market impact, leverage, and capital gains tax on strategy performance.

Table 4. Evaluation of Empirical Results



Thank You

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